

Loops, Goals, and Iterations

Lecture 17 — Agents, Skills, and Hooks

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Placeholder

Why one-shot prompting is fragile; the goal-directed loop (observe $\rightarrow$ propose $\rightarrow$ act $\rightarrow$ evaluate $\rightarrow$ revise); loops vs. repetition; Claude Code as the substrate; when *not* to use a loop.

Placeholder

Concrete, checkable, decomposable goals; success criteria, stopping rules, failure modes; goals as files; bad vs. good goals.

Placeholder

Roles (worker, critic, validator, planner, refactorer); single- vs. multi-agent loops; handoffs; agents improve project *state*.

Placeholder

Skills as reusable operational knowledge; skills $\approx$ functions; skills encode conventions and reduce entropy.

Hooks as Guardrails and Feedback Channels

Placeholder

Pre/post hooks; the hook catalogue; measurable feedback; trustworthy validators; failure hooks.

Designing an Iteration Loop

Placeholder

The seven-step template; pseudocode; state and context across iterations; stopping criteria; the economics of iteration.

Example 1: Derivative-Free Root Finding

Placeholder

Find a zero of $f(x) = x^3 - x - 2$ without derivatives; goal file, evaluator skill, success hook; iteration trace from $x = 0$; final report.

Example 2: Credit-Risk Stress-Testing Loop

Placeholder

Credit-committee memo (PD, LGD, expected loss, recommendation); combined macro shock; agent ensemble; skills; hooks; eleven-iteration narrative; final table.

Placeholder

Human-in-the-loop and escalation; sign-off and model-risk governance; reproducible audit trails.

Placeholder

Infinite loops; fake convergence; goal drift; Goodhart overfitting; excessive delegation; warning signs and remedies.

Placeholder

Planner-executor-critic; generate-test-revise; scaffold-draft-audit-repair; data-pipeline build-validate-document; credit-model build-stress-test-red-team-finalize.

Summary and Takeaways

Placeholder

Iteration helps only when feedback is informative; agents specialize, skills reuse, hooks enforce; finance needs explicit goals, validation, stress testing, and skepticism.